Microsoft

MICROSOFT × GLOBAL DARUSSALAM ACADEMY

AI for Learning & Teaching



Yogyakarta · Microsoft Indonesia



Microsoft Indonesia

SPECIALISATION

Big Data | Analytics | AI

INTRODUCTION

Hello, I'm

Edhot Purwoko

Sr Technology Specialist

20+ **Years in the tech industry**
5 years at Microsoft · previously Nokia & Huawei

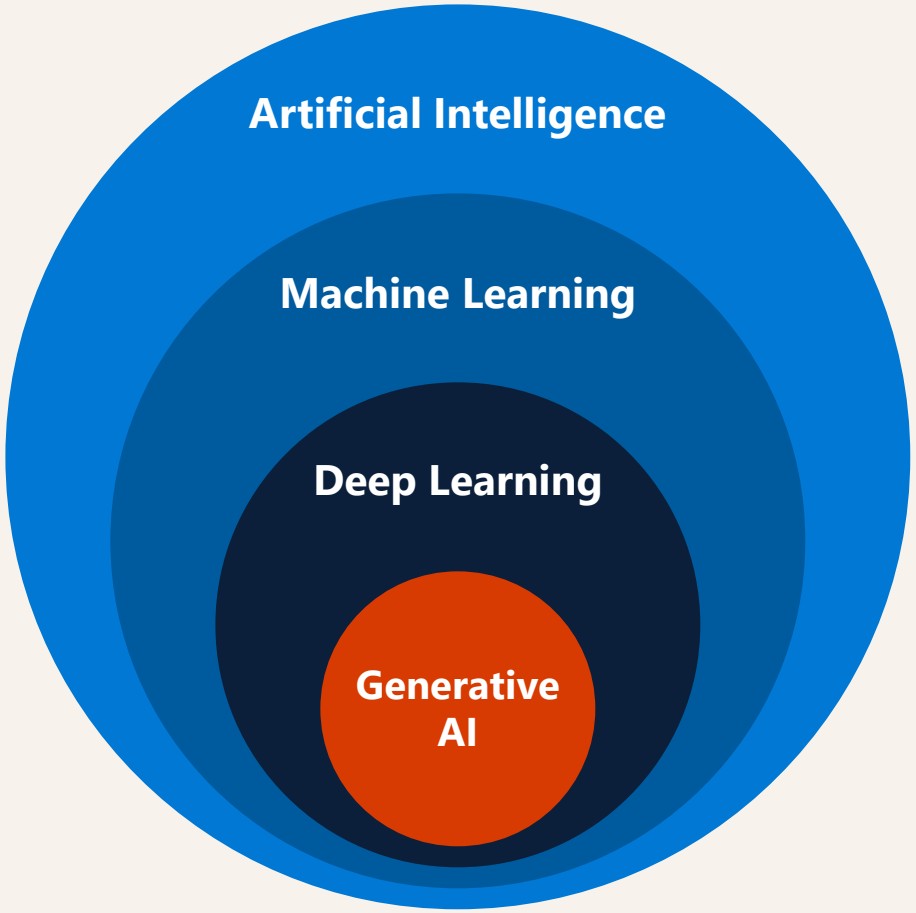
EDUCATION

Master's · Information Technology
Universitas Indonesia (University of Indonesia)

Bachelor's · Informatics Engineering
Universitas Islam Indonesia (Islamic University of Indonesia)


AI is transforming the world


Data is the fuel
that *powers* AI



1956

Artificial Intelligence

The field of computer science that builds intelligent machines that mimic human thinking.

1997

Machine Learning

Machines learn patterns from data to make predictions — e.g. video recommendations.

2017

Deep Learning

Layered neural networks that recognise images, sound and language — e.g. face recognition.

2021

Generative AI

Creates new text, images and audio from a prompt — like Copilot and Designer.

Each smaller circle is a subset of the larger one: Generative AI ⊂ Deep Learning ⊂ ML ⊂ AI.

How does generative AI answer?

1 Learns from lots of text

A large language model (LLM) is trained on huge amounts of text, so it recognises language patterns.

2 Predicts the next word

When answering, the LLM builds the answer piece by piece (tokens) from the most likely pattern.

3 Can be wrong, so check

Answers can sound convincing but be wrong. Double-check with books or trusted sources.

FROM PROMPT TO RESULT



SCHOOL EXAMPLES

- "Summarise the Mataram Kingdom chapter in 5 points." → summary
- "Give 3 short-story ideas set on Malioboro." → creative ideas
- "Make this paragraph simpler." → new text

Source: Microsoft Learn – LLM Fundamentals; Introduction to generative AI and agents; Microsoft Support – Copilot for young people

OVERVIEW · THE KEY TO SUCCESS

A good prompt has 4 parts

1 Goal	2 Context	3 Source	4 Expectations
What do you want?	**For whom?**	**Which material?**	**What format?**
"Create 5 practice questions"	"for grade 8 students just starting Pythagoras"	"based on the material file I uploaded"	"as a table, with an answer key"

FULL PROMPT EXAMPLE

"Create 5 practice questions for grade 8 students just starting Pythagoras, based on the material file I uploaded, as a table with an answer key."

Source: Microsoft Support – Get started writing prompts in Microsoft Copilot (Goal, Context, Source, Expectations)

Generative AI Adoption Patterns

Wave 1 and Wave 2 – 2023 to 2024, moving to scale in 2025

Wave 1

Use Cases Inside the Organization

Internal Copilot deployments

Wave 2

Infusing Generative AI to your products

+

Customer Experience

Wave 3

Agentic AI

Autonomous and Semi-Autonomous agents

Wave 4

Artificial General Intelligence

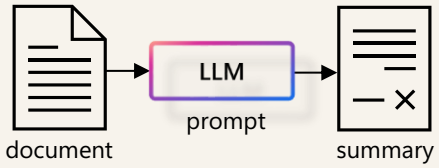
New Product Offerings

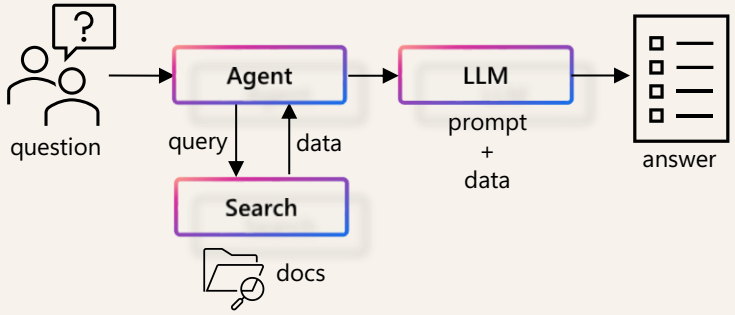
Human 'in the loop' review of generative content

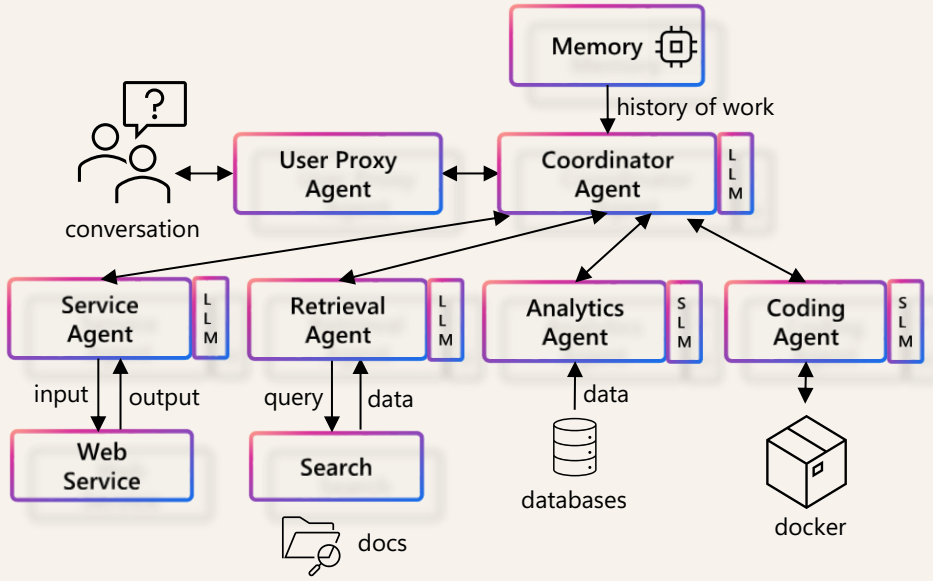
AI augmenting workgroups

Human Supervision of multiple agents

Evolution of LLM-based Solutions







No Agent

Very narrow one shot task

Ex: log to JSON

Single Agent

Very clearly scoped iterative task

Ex: providing an answer with supporting evidence to a complex question

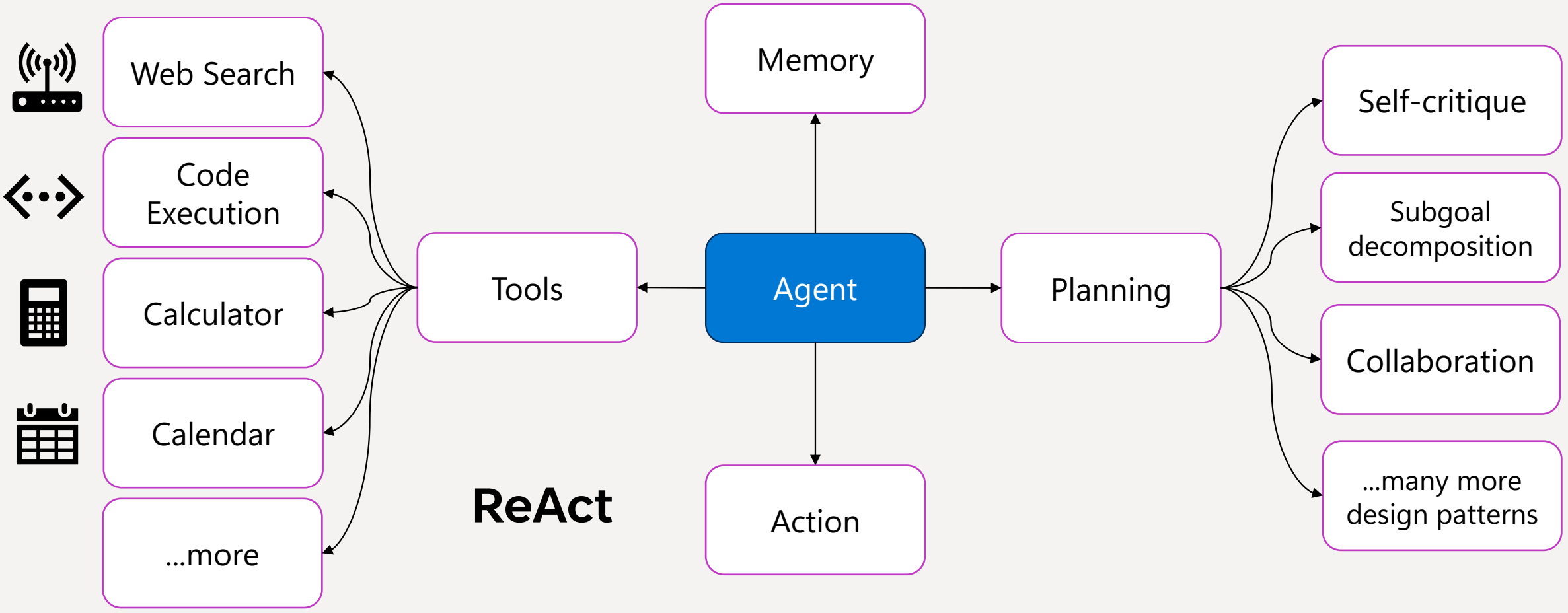
Multi-agent Systems

Wide scope complex use case requiring diverse skills

Ex: Propose 2 Instagram marketing campaigns including assets that would leverage the top 2 recent trends in our past quarter US Sales to boost our mailing list user base and predict the impact of each campaign

VALUE

Agentic AI capabilities



Microsoft

HANDS-ON SESSION

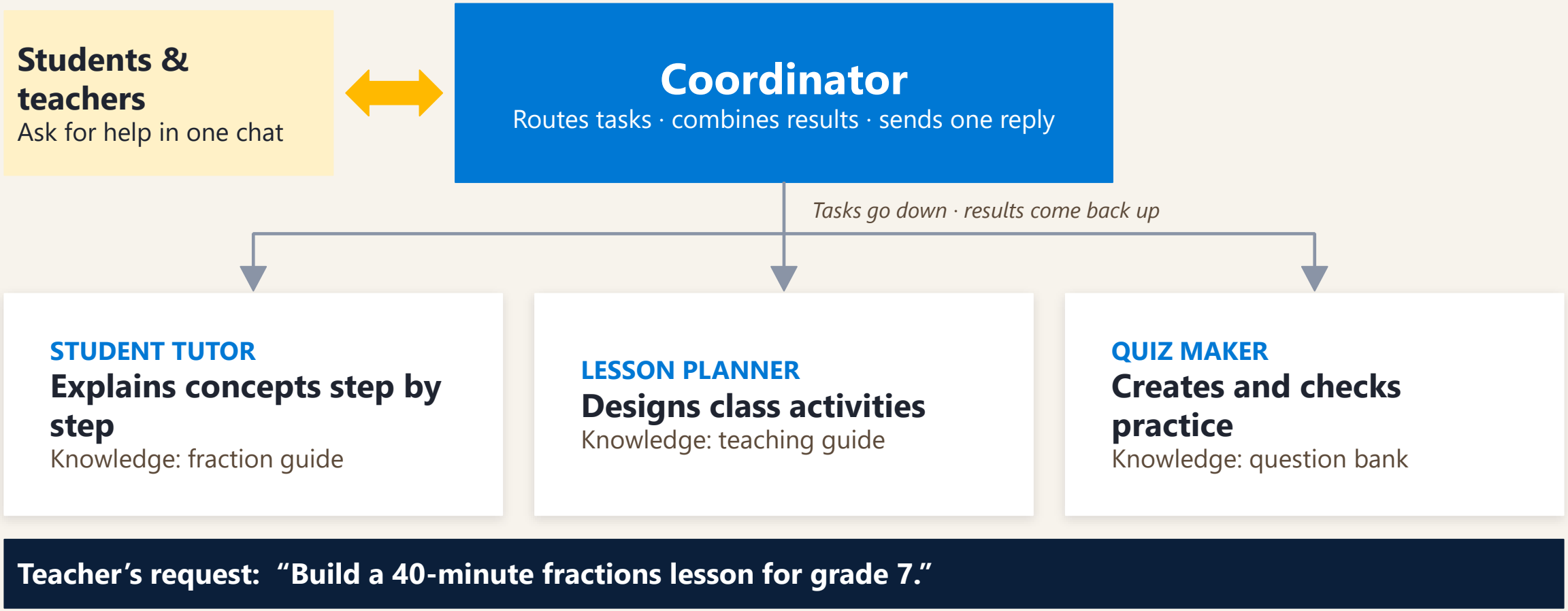
Let's start the demo



MULTI-AGENT DEMO · MICROSOFT COPILOT STUDIO

One conversation, three specialist agents

Team “Sahabat Belajar”: a coordinator routes the request, combines results and sends one reply.



Source: Microsoft Learn – Copilot Studio: Add other agents (child agents); demo configuration

HANDS-ON AGENDA

Three demos, one goal: better learning

1

DEMO 1 · AI

Study Buddy

Learn with AI, not cheat with it

Microsoft Copilot – Copilot Learn

For students

2

DEMO 2 · AI

Creative Studio

From idea to visual work

Microsoft Copilot + Microsoft Designer

For students

3

DEMO 3 · AI

Teaching Assistant

Learning materials in 5 minutes

Microsoft Copilot (personal account)

For teachers

PREPARATION

Set up access before the demo

DEMO 1 · AI
Study Buddy

OPEN IN

Copilot → More → Learn → Guided learning

copilot.cloud.microsoft

NOTE

Make sure the “Personal” label shows in the left panel.

DEMO 2 · AI
Creative Studio

OPEN IN

Microsoft Designer website or app

designer.microsoft.com

NOTE

Limited free quota; personal, non-commercial use.

DEMO 3 · AI
Teaching Assistant

OPEN IN

Copilot → chat, or More → Learn

copilot.cloud.microsoft

NOTE

Use copy-ready prompts; the Teach feature is for school accounts only.

ACCOUNT FOR ALL DEMOS

Personal Microsoft account (free)

Don't have one? Sign up at signup.live.com

DAY-BEFORE CHECK

Make sure everyone can sign in, open both links above on the school Wi-Fi, and prepare 1–2 material files.

Source: Microsoft Learn – Manage Copilot Chat; Microsoft Support – Copilot Learn; Microsoft Designer FAQ

1

DEMO 1 · FOR STUDENTS

AI Study Buddy

Learn with AI, not cheat with it

Guided Learning in Copilot Learn walks students through step-by-step questions, then quizzes and flashcards. Students do the thinking; AI does the coaching.

Solution: Microsoft Copilot – Copilot Learn



DEMO 1 · AI STUDY BUDDY

Follow these 6 steps

1. Open copilot.cloud.microsoft and sign in with a personal Microsoft account.
2. In the left panel choose More → Learn, then click Guided learning.
3. Optional: use Upload files to learn from your own material.
4. Type a topic and ask for the Socratic method — use the sample prompt on the right.
5. Answer the AI's questions in your own words; ask for hints, not answers.
6. Go back to Learn and pick Quiz or Flashcards to test yourself.

SAMPLE PROMPT

"Teach me photosynthesis using the Socratic method. Don't give me the answer straight away — ask one question at a time, give hints when I'm wrong, then make a 5-question quiz."

Swap the topic:

- Linear equations in one variable
- The 1928 Youth Pledge
- History of the Yogyakarta Palace

Source: Microsoft Support – Use a screen reader to explore and use Copilot Learn

DEMO 1 · AI STUDY BUDDY

Then test your understanding

Flashcards

Study cards from a chat or uploaded file

Quiz

Quizzes from a topic or material; resume and track progress

Practice test

Ask in chat: “Make a practice test from this material.”

Homework help

Snap a problem; Copilot guides you step by step

FOLLOW-UP PROMPTS — TYPE IN COPILOT LEARN

“Make 10 flashcards from the material I uploaded.”

“Give me a 5-question quiz and explain each of my answers.”

“Explain it again with everyday examples from Yogyakarta.”

Source: Microsoft Support – Use a screen reader to explore and use Copilot Learn (Quizzes, Flashcards, Practice Tests)

2

DEMO 2 · FOR STUDENTS

AI Creative Studio

From idea to visual work

Students create a campaign poster: write a prompt, generate an image, refine it, then lay out a presentation-ready poster.

Solution: Microsoft Copilot + Microsoft Designer



DEMO 2 · AI CREATIVE STUDIO

From prompt to poster in 5 steps

1 Sign in	2 Prompt	3 Tingkatkan	4 Edit	5 Bagikan
Open the Designer website or the Designer mobile app and sign in with a personal Microsoft account (free).	Choose create image, then write your poster idea using the prompt formula on the next slide.	Click Enhance prompt so AI adds detail, edit as needed, then select Generate.	Pick the best image, click Edit to open the canvas, then add a title with the Text menu.	**3 Enhance**

IMPORTANT FOR PARTICIPANTS

Text inside AI images may be misspelled or in the wrong language — write the poster title with the Text menu on the Designer canvas.

Images from Designer's AI features carry content credentials (C2PA standard) that mark them as AI-made.

5 Share

DEMO 2 · AI CREATIVE STUDIO

Campaign poster prompt templates

PROMPT FORMULA

1 Subject
what or who appears
2 Style
modern batik, watercolour, flat
3 Colour
two or three main colours
4 Mood
cheerful, calm, futuristic
5 Title space
empty area for text

Refine: “make it brighter”, “switch to a watercolour style”.

Plastic-Free Yogyakarta

“Modern batik illustration poster: Tugu Yogyakarta and a clean river, residents sorting plastic waste, green and brown, empty space for a title.”

Save the Earth

“Watercolour illustration: the Earth held by children's hands, green forests and blue seas, warm light, empty space at the bottom for a slogan.”

Digital Safety for Students

“Flat illustration: a student on a laptop protected by a glowing padlock shield, blue and white, empty space on the left for a title.”

Future School in 2035

“Bright digital art: a green school in Yogyakarta in 2035, joglo roofs with solar panels, students learning in a garden, space for a title.”

Tip: add the poster title with the Text menu on the Designer canvas, not inside the image prompt

3

DEMO 3 · FOR TEACHERS

AI Teaching Assistant

Learning materials in 5 minutes

With a personal Microsoft account, teachers use Microsoft Copilot to create lesson plans, questions, rubrics, flashcards and differentiated materials.

Solution: Microsoft Copilot + Copilot Learn (personal account)



DEMO 3 · AI TEACHING ASSISTANT

One assistant for teachers' needs

PLANNING

Lesson Plan

Via chat or Learn → Create a page

ASSESSMENT

Quiz

Questions via chat, practice via Learn → Quiz

ASSESSMENT

Rubric

Ask for clear levels and criteria

STUDY TOOLS

Flashcards

Practice with flashcards from text or a file

ADAPT CONTENT

Differentiation

Adapt tasks for different ability levels

ADAPT CONTENT

Reading level

Simplify or stretch text to match ability

ADAPT CONTENT

Real-world examples

Add context and real-world examples

ADAPT CONTENT

Standards

Ask for alignment with the national curriculum

Source: Microsoft Support – Copilot Learn; Get started writing prompts in Microsoft Copilot

DEMO 3 · AI TEACHING ASSISTANT

Build a lesson plan in 6 steps

1 Open Copilot

Open copilot.cloud.microsoft and sign in with a personal Microsoft account.

2 Choose a mode

Regular chat, or More → Learn → Create a page.

3 Add class details

State the subject (Math), grade (8), duration and language in the prompt.

4 Write the goals

Write learning goals and the output format: goals, activities, assessment.

5 Refine

Ask for revisions: “shorter”, “add group work”, “align with Kurikulum Merdeka”.

6 Save

Copy it to a document, review it, then move on to questions and a rubric.

Source: Microsoft Support – Copilot Learn (Start a new doc with AI); Get started writing prompts in Microsoft Copilot

DEMO 3 · AI TEACHING ASSISTANT

Example: 10 grade 8 math questions

PROMPT IN MICROSOFT COPILOT (PERSONAL ACCOUNT)

"You are a junior high math teacher. Create 10 multiple-choice questions on the Pythagorean theorem for grade 8: 4 easy, 4 medium, 2 hard. Use everyday contexts in Yogyakarta. Include an answer key and step-by-step explanations."

Expected output:

- 10 questions at three difficulty levels
- An answer key for every question
- Step-by-step explanations

Rubric

"Create a 4-level rubric for a Pythagoras problem-solving task."

Differentiation

"Create easier versions of questions 1–5 for students who need support."

Interactive quiz

Students practise the same topic via Learn → Quiz.

Teacher stays the reviewer: check the answer key before sharing.

Source: Microsoft Support – Get started writing prompts in Microsoft Copilot; Copilot Learn (Quizzes)

RESPONSIBLE AI

Four good habits when using AI

AI helps. Humans keep thinking.

We practise these habits in all three demos today.

Students do the thinking

Guided Learning coaches through questions, not answers.

Check facts and sources

Copilot includes citations; double-check with your textbook.

Be honest about AI work

Designer AI images carry content credentials; say when work was made with AI.

Teachers stay the reviewer

Review lesson plans, questions and answer keys before use.

Source: Microsoft Support – Copilot Learn; Copilot for young people; Microsoft Designer FAQ

Microsoft

HANDS-ON SESSION

AI for Accessibility



AI FOR ACCESSIBILITY · MICROSOFT × UNIVERSITAS INDONESIA

UI students build AI for inclusion

Microsoft's annual AI for Accessibility (AI4A) Hackathon challenges young innovators in Southeast Asia to solve real problems faced by people with disabilities. The 2025 edition was co-hosted with UI's Faculty of Engineering.

6th
year of the AI4A Hackathon

46
teams competed

10
grand finalists

1ST PLACE

NeuroBuddy

Team The Leporidaes

A children's game with a rabbit mascot; AI evaluates play to flag early signs of dyslexia and ADHD.

Built on Azure AI services

2ND PLACE

Pintaru

Team UINNOVATORS

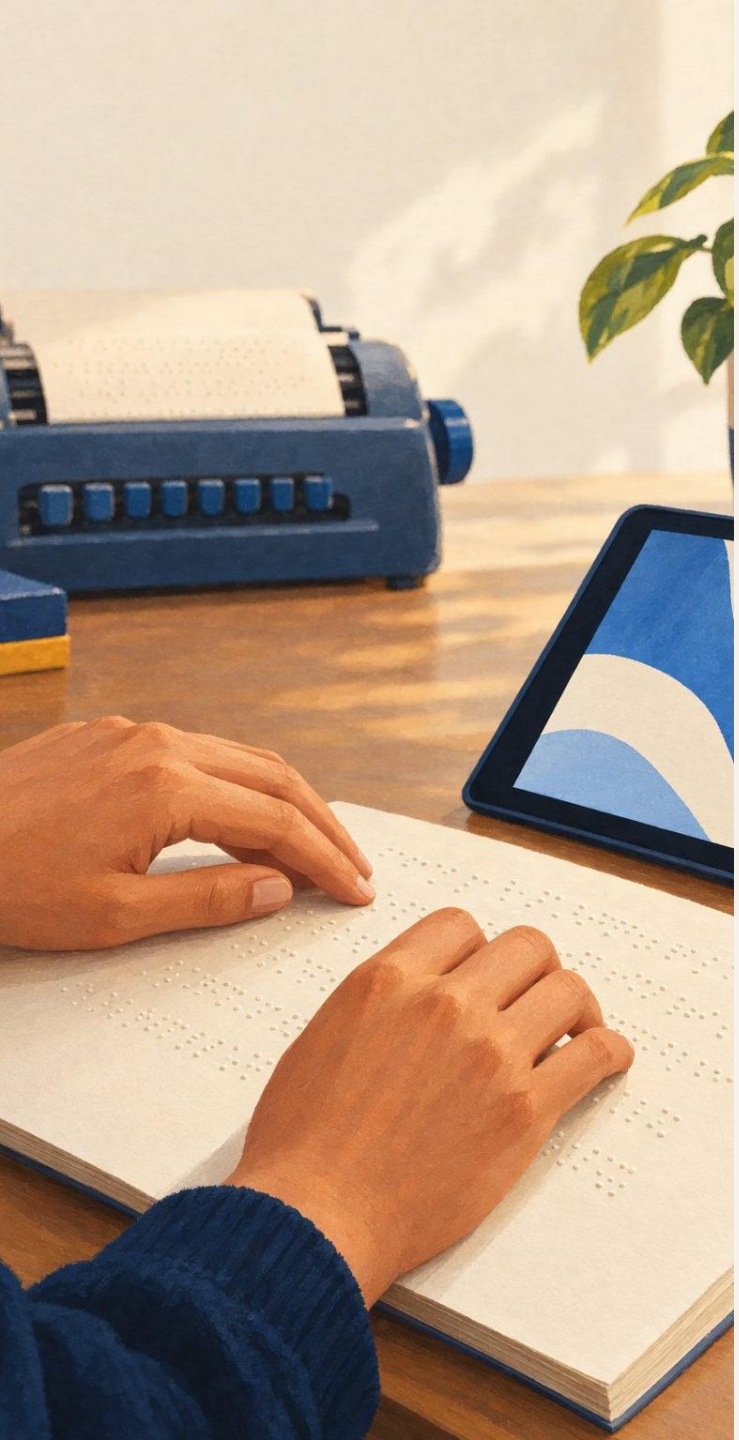
An adaptive digital book that adjusts font size, spacing and visuals for students with dyslexia.

Azure OpenAI · AI Search · Speech

Grand-final jury: Rahma Utami (Suarise) · F. Astha Ekadiyanto (FT UI) · Edhot Purwoko (Microsoft)

Source: Microsoft Indonesia News Center (26 Jun 2025); Universitas Indonesia (3 Jul 2025)





AI FOR ACCESSIBILITY · MICROSOFT × YAYASAN MITRA NETRA

Arabic Braille, powered by AI

Yayasan Mitra Netra empowers blind people in Indonesia. Its Arabic Braille Converter lets teachers who can't read Braille and blind students work on the same Arabic and math lessons.

Photo or scan
Arabic text with harakat

→

Azure OpenAI
reads letters + harakat

→

Arabic Unicode
editable digital text

→

Indonesian Braille
and back to text

2022
5th place, Microsoft AI4A Hackathon Asia Pacific

APR 2024
Showcased at Microsoft Build: AI Day Jakarta

DEC 2024
Featured in Microsoft Indonesia's "AI for Indonesia"

2025
Adds math formulas and a Braille keyboard

IMPACT IN CLASSROOMS
Used in special schools (SLB) and inclusive schools with Braille printers: teachers type in Word, convert and print Braille; blind students type in Braille and teachers read it as text.

Sources: Microsoft Indonesia News Center (Dec 2024); Bisnis.com; kumparan (Oct 2025)

MICROSOFT × EDUCATION

Free Microsoft programs for education

What is free for schools, teachers and students, who can join, and where to start.

Schools | Teachers | Students

Global Darussalam Academy · Yogyakarta



FOR SCHOOLS

Office 365 A1: free for eligible schools

US$0

Office 365 A1 for Education

Microsoft donates A1 to eligible institutions, for educators and students, at no cost.

Start: microsoft.com/education

PRODUCTIVITY

Office on the web + Teams

Word, Excel, PowerPoint, OneNote, Forms and Teams in the browser; 50 GB mailbox.

AI

Microsoft 365 Copilot Chat

Secure AI chat with enterprise data protection, plus web-grounded agents.

LEARNING

Learning Accelerators

Reading Progress & Coach, Search Coach, Speaker Coach, Math progress, Reflect.

INCLUSION

Accessibility built in

Immersive Reader, Translator, live captions and voice typing.

Compare plans: microsoft.com/en/education/products/microsoft-365/compare-microsoft-365-education-plans



FOR TEACHERS

Free AI training for teachers

NEW · LAUNCHED JAN 2026

Microsoft Elevate for Educators

Free courses, live sessions and AI simulations in 13+ languages, a free Elevate Educator credential (with ISTE + ASCD) and global communities.

→ **aka.ms/MicrosoftElevateforEducators**

COURSES & EXPERT RECOGNITION

Microsoft Learn Educator Center

Free self-paced training for educators. The former MIE Expert community now continues inside Microsoft Elevate Educators.

→ **learn.microsoft.com/training/educator-center/programs**

HIGHER-ED & STEM SECONDARY

Microsoft Learn for Educators (MSLE)

Free official Microsoft curriculum and faculty training to teach AI and cloud skills toward Microsoft credentials.

→ **learn.microsoft.com/training/educator-center/programs/msle/overview**

FOR STUDENTS

Free tools and a world stage

AGE 13+ · VERIFIED STUDENTS

GitHub Student Developer Pack

Free GitHub Pro, Codespaces and Copilot Student, plus partner tools.
→ **github.com/education/students**

FULL-TIME UNIVERSITY STUDENTS

Azure for Students

US$100 credit, no credit card; renew yearly while studying.
→ **azure.microsoft.com/free/students**

EVERYONE · PERSONAL ACCOUNT

Microsoft Learn

Free AI and cloud learning paths; progress and XP are saved.
→ **learn.microsoft.com/ai**

18+ · STUDENT-LED STARTUP

Imagine Cup 2027

Student Prize: US$25K + US$250K credits. Apply by 28 Feb 2027.
→ **imaginecup.microsoft.com**

FREE CREDENTIALS · STUDENTS & EDUCATORS

Microsoft Applied Skills: free, lab-based AI and cloud credentials.
→ **learn.microsoft.com/credentials/applied-skills**

Sources: GitHub Docs; Azure for Students; Imagine Cup 2027 Official Rules; Microsoft Applied Skills



KEEP LEARNING · FREE RESOURCES FROM MICROSOFT

AI learning map: from beginner to advanced

LEVEL 1 Beginner

FOR: STUDENTS & TEACHERS

Microsoft Learn: AI Fluency
7 modules, ±4.5 hrs: AI basics, generative AI, prompting, responsible AI.

Minecraft Education: Hour of AI
"The First Night" & "Reed Smart: AI Detective", free via demo lessons.

LEVEL 2 Practitioner

FOR: TEACHERS

AI for educators
4 modules, ±4 hrs: LLMs, prompting, AI in class.

AI-powered teaching with Microsoft Education
9 modules, ±6 hrs: from AI basics to assessment.

LEVEL 3 Intermediate

FOR: HIGH SCHOOL & UNIVERSITY

AI concepts for developers
7 modules: NLP, vision, speech, RAG; AI-901 exam prep.

GitHub: ML & AI for Beginners
26 + 24 free lessons, with Indonesian translations.

LEVEL 4 Advanced

FOR: UNIVERSITY STUDENTS & DEVELOPERS

Generative AI for Beginners
21 lessons: advanced prompting, RAG, AI agents, fine-tuning.

Azure for Students
USD100 credit for 12 months, no credit card (18+, school email).

Source: Microsoft Learn – AI learning hub & Training; GitHub microsoft (AI/ML/Generative AI for Beginners); Minecraft Education – Hour of AI; Azure for Students offer

KEEP LEARNING · START TODAY

How to make the most of Microsoft Learn

4 STEPS TO START

1. Open learn.microsoft.com/ai and sign in with a personal Microsoft account — your progress and XP are saved.
2. Pick a learning path for your level on the map; start at Beginner.
3. Do the exercises and module assessments; save favourite paths to Collections.
4. Ready to certify? The material is free; the exam (e.g. AI-901) is paid.

PROGRAMME IN INDONESIA

Microsoft Elevate Indonesia

Formerly elevAIte Indonesia. Free AI training with Komdigi and partners: >1.2 million participants since Dec 2024, targeting 500,000 certified talents in 2026.

For educators: AI Teaching Power & AI Coding for Educators.

This week's challenge

Students: finish the “Explore AI basics” module + 1 Hour of AI.
Teachers: the “Empower educators to explore the potential of AI” module (±1 hr).

Source: Microsoft Learn – AI learning hub; Microsoft Source Asia – Microsoft Elevate Indonesia (Nov 2025)

START HERE · OFFICIAL LINKS

Official Microsoft links to bookmark

SCHOOLS	**Office 365 A1 – compare Microsoft 365 Education plans**	microsoft.com/en/education/products/microsoft-365/compare-microsoft-365-education-plans
TEACHERS	**Microsoft Elevate for Educators**	aka.ms/MicrosoftElevateforEducators
TEACHERS	**Microsoft Learn Educator Center – educator programs**	learn.microsoft.com/training/educator-center/programs
TEACHERS	**Microsoft Learn for Educators (MSLE)**	learn.microsoft.com/training/educator-center/programs/msle/overview
STUDENTS	**GitHub Education for students (Student Developer Pack, Copilot Student)**	github.com/education/students
STUDENTS	**Azure for Students**	azure.microsoft.com/free/students
STUDENTS	**Imagine Cup 2027 – applications open until 28 Feb 2027**	imaginecup.microsoft.com
EVERYONE	**Microsoft Learn – AI learning hub**	learn.microsoft.com/ai
EVERYONE	**Microsoft Applied Skills – free lab-based credentials**	learn.microsoft.com/credentials/applied-skills

All programs re-checked 8 Nov 2026; eligibility varies by country, age and school status.

Microsoft

Matur nuwun! (Thank you!)

Thank you — let's learn, create and teach with AI.

Q&A and free practice

Workshop Materials

Scan for PDF & tutorials

github.com/edhotp/
MATERI_WORKSHOP_GDA